

⚠️ **EARLY-VERSION TEST MOCK-UP, ILLUSTRATES THE IDEA AND THE GOAL ONLY · NOT A DESIGN · NOT FOR CONSTRUCTION · EVERYTHING SUBJECT TO COMPLETE CHANGE** ⚠️

VERY EARLY CONCEPT ILLUSTRATION, NOT A DESIGN DOCUMENT. The complete BRONTE Chimborazo design map in one sheet: route (real SRTM 30 m terrain), both build phases, the tunnel & tube section with the 360° drive ring, and engineering charts for both. Tube architecture: full-circumference stator, levitation, centering, and propulsion from all sides, sized for a Ø3.2 m payload. All dimensions speculative and unvalidated.

ANNEX B

BRONTE, Chimborazo Design Map

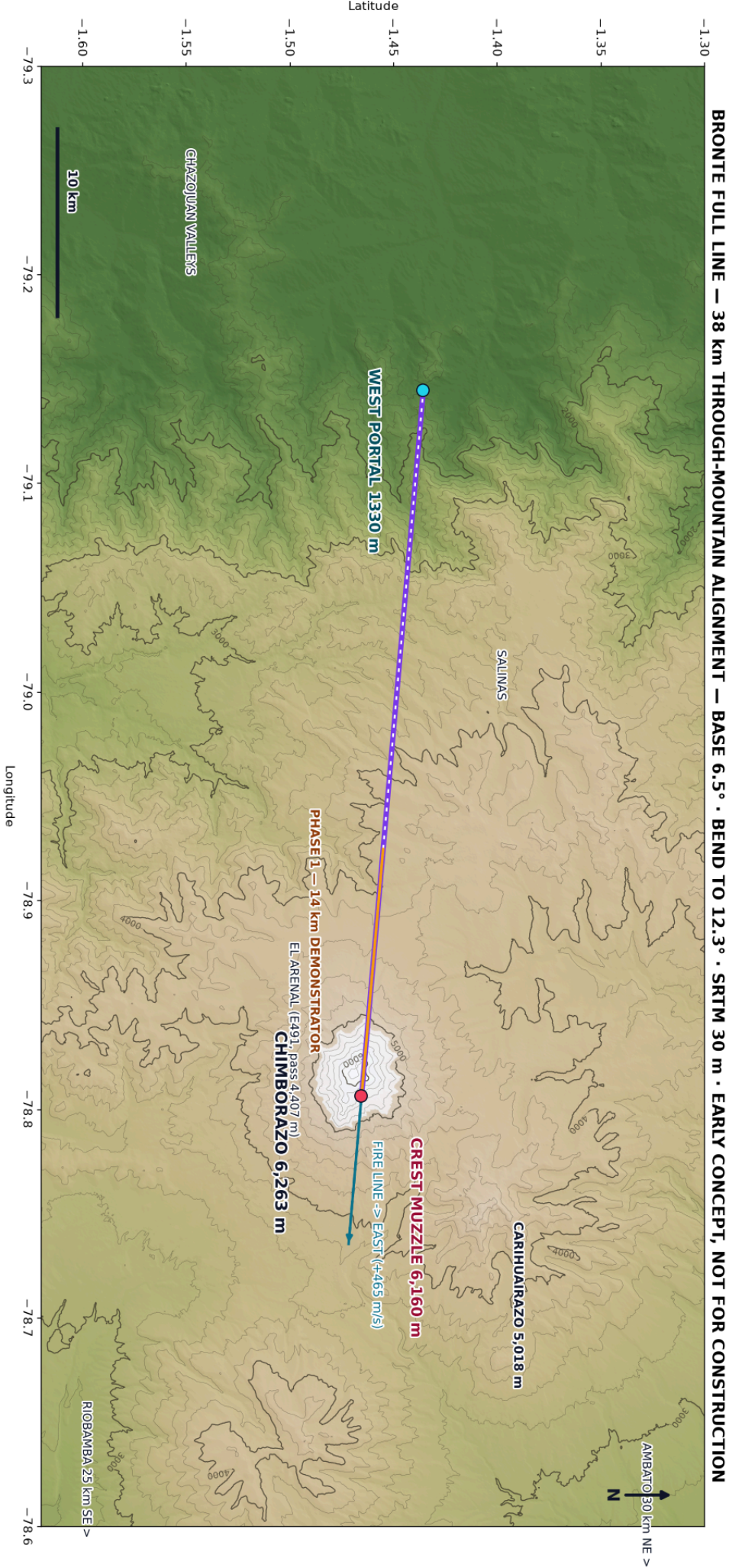
KER-BRO-TRK-001 · REV N · EARLY-VERSION TEST MOCK-UP · NOT FOR CONSTRUCTION

WHAT IS INSIDE

- ▶ Panel A, Plan view: one line, two phases (real SRTM 30 m terrain)
- ▶ Panel B, Phase 1 (build first): 14 km, portal 2,920 m → tunnel muzzle 5,900 m, straight 12.3° (true 1:1)
- ▶ Panel C, Full line (~21.3 km growth target): single straight bored tunnel @ 12.3°, no bend (true 1:1)
- ▶ Panel D, Tunnel & tube typical section: the 360° drive ring, wide bore for Ø3.2 m payload
- ▶ Panel E, External bridge section · ASTRAPE front view · ASTRAPE in the tube
- ▶ Charts, systems, route engineering, tunnel engineering

Each large panel is on its own page, rotated to read at full size, turn the page (or your screen) a quarter-turn clockwise.

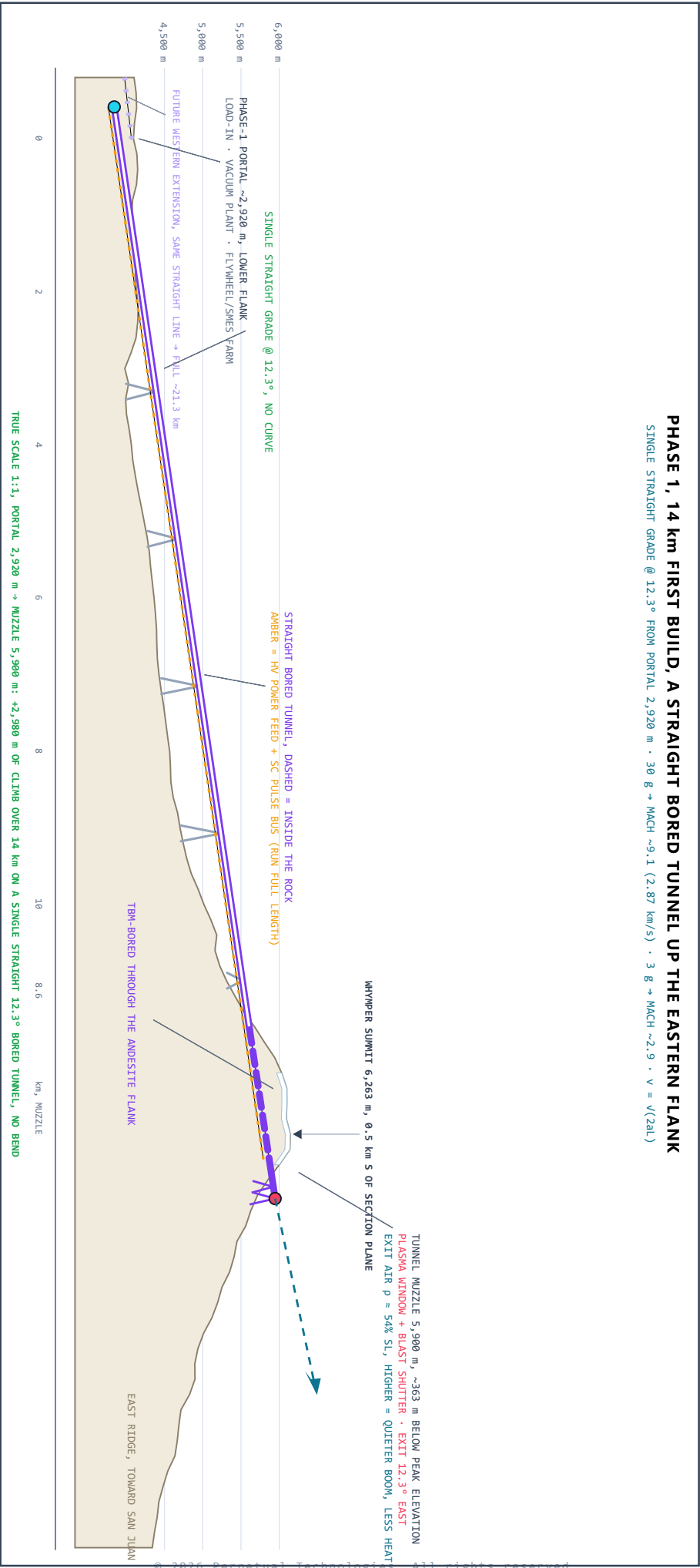
PANEL A, PLAN VIEW: ONE LINE, TWO PHASES (SRTM 30 m)



PANEL B, PHASE 1 (BUILD FIRST): 14 km, PORTAL 2,920 m → TUNNEL MUZZLE 5,900 m, STRAIGHT 12.3° (TRUE 1:1)

PHASE 1, 14 km FIRST BUILD, A STRAIGHT BORED TUNNEL UP THE EASTERN FLANK

SINGLE STRAIGHT GRADE @ 12.3° FROM PORTAL 2,920 m · 30 g → MACH ~9.1 (2.87 km/s) · 3 g → MACH ~2.9 · v = v(2a)



PANEL C, FULL LINE (THE GROWTH TARGET): ~21.3 km, STRAIGHT BORED TUNNEL @ 12.3°, NO BEND (TRUE 1:1)

FULL LINE, WEST VALLEYS → STRAIGHT BORED TUNNEL @ 12.3° → TUNNEL MUZZLE 5,900 m

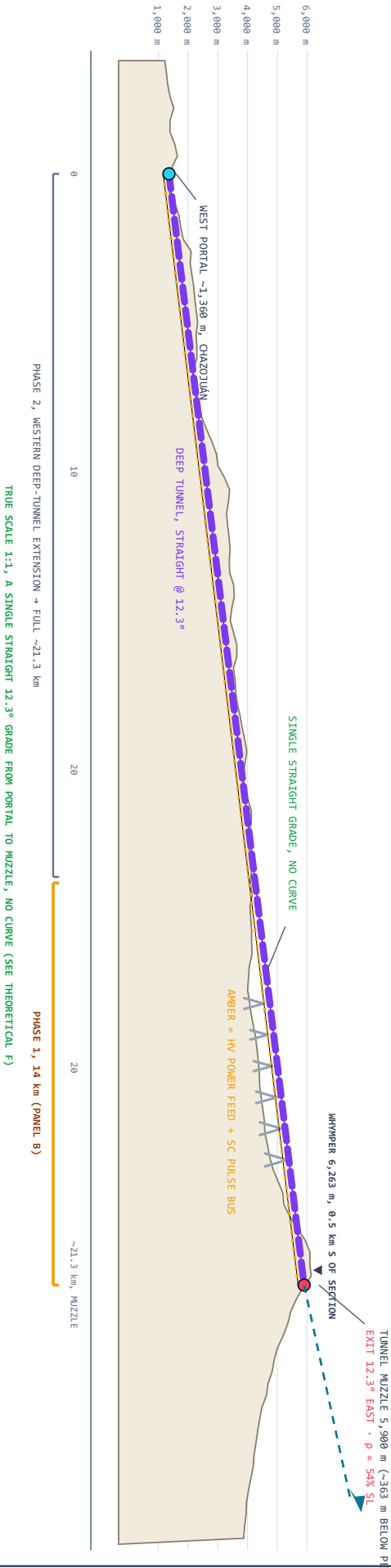
WEST PORTAL ~1,360 m (CHAZOJUAN) · TUNNEL MUZZLE 5,900 m (PEAK 6,263 m) · EXIT 12.3° EAST · TRUE SCALE 1:1

GENERAL NOTES

1. HIGH-SPEED TRACK CANNOT CURVE, IT IS A SINGLE STRAIGHT LINE AT 12.3°. NO CURVE ⇒ ZERO LATERAL G (A CURVE WOULD BE 13+ G).
2. WHY ALTITUDE: EXIT AIR @ 5,900 m IS $\rho = 54\%$ SL ($\rho = 47\%$) · EVERY METER HIGHER MEANS A WEAKER GROUND-LEVEL BOOM, LESS DRAG, LESS HEATING.
3. PRICE OF THE GREST EXIT: EL ARENAL PIERS REACH ~590 m (MILLAU RECORD 343 m) · A-FRAME PIERS FOR SEISMIC STIFFNESS, OPEN ISSUE.
4. WEST DEEP TUNNEL 24 km, COVER $\pm \sim 880$ m (GOTTHARD: 2,300 m) ·
5. SEISMIC -0.4 G (PALATANGA) · VOLCANO DORMANT (~550 CE) · RESERVE LEGAL STATUS OPEN · FIRES EAST: +465 m/s ROTATION BONUS FREE.
6. REV H CORRECTION: 6,255 m MUZZLE RETRACTED (NEEDED ~520 m MAST).

PERFORMANCE, BASELINE 15 t · MAX GROSS 20 t

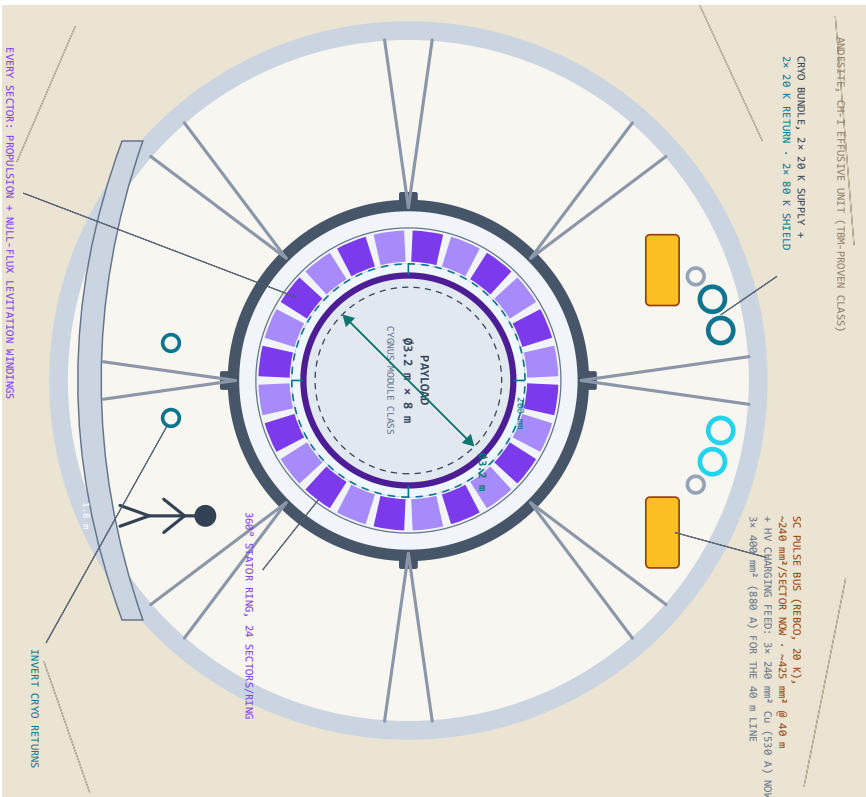
FULL (21.3 km), 30 G + 3.54 km/s · MACH ~11.2
FULL, 3 G HUMAN + 1.12 km/s · MACH ~3.5
PHASE 1 (14 km), 30 G + 2.87 km/s · MACH ~9.1
PHASE 1, 3 G + 0.91 km/s · MACH ~2.9
LATERAL G: 0, STRAIGHT TRACK, NO BEND ($v=\sqrt{2aL}$)
MUZZLE 5,900 m (JUST UNDER 6 km) · ~363 m BELOW PEAK
TIME IN THE TUBE (FROM REST TO EXIT):
30 G, 12.0 s FULL · 9.8 s P1 3 G, 38 s FULL · 31 s P1
 $v = \sqrt{(2aL)}$: $a = 30 \text{ G} = 294 \text{ m/s}^2$, $L = 21.315 \text{ m} \Rightarrow 3,540 \text{ m/s}$
AT 20 t MAX: THRUST 5.9 MN · 34.8 MN · PEAK 20.8 GN



PANEL D, TUNNEL & TUBE TYPICAL SECTION, 360° DRIVE RING, WIDE BORE FOR Ø3.2 m PAYLOAD

INSIDE THE TUNNEL, FULL-CIRCUMFERENCE STATOR: LEVITATE, CENTER, AND PROPEL FROM ALL SIDES

Ø12.0 m TBM BORE · Ø6.0 m VACUUM VESSEL · 360° STATOR RING Ø4.0-5.2 m · Ø4.0 m FLIGHT BORE · Ø3.6 m VEHICLE · GAP 200 mm ALL ROUND · DIAMOND TO SCALE



WHY THE RING IS 360° ON A STRAIGHT TRACK

A STRAIGHT TRACK HAS NO CURVE, SO NO 1/6 SIDE-LOAD. THE FULL-CIRCUMFERENCE STATOR STAYS FOR TWO REASONS:
1. DISTRIBUTED FULL-HULL THRUST, DRIVER ALONG THE WHOLE SLEEVE, SO THE POD NEVER BODIES AHEAD, BLOCKING LOAD.
2. PRECISE MAGNETIC CENTERING OF A MACH-11 VEHICLE IN A 200 mm VACUUM GAP, NO CONTACT, NO FLOOR GUIDANCE, GRAVITY (1 g) + ALIGNMENT ARE THE ONLY LATERAL LOADS NOM.

SECTION STACK, SIZED FROM THE PAYLOAD OUT

Ø12.0 m TBM BORE, PRECEDENT: BERTHA 17.45 m · CCS 9.04 m
0.32 m SEGMENTAL LINER + GROUT
Ø6.0 m VACUUM VESSEL, 40 mm 316L + RINGS @ 2.5 m
SIZED BY EXTERNAL CRUSH (BUCKLING), NOT BURST
Ø4.0-5.2 m STATOR RING, 24 SECTORS × HOOPS EVERY 0.6 m
~37,000 COIL HOOPS ON THE FULL ~21.3 km LINE
Ø4.0 m FLIGHT BORE · ~10 Pa · GAP 200 mm ALL ROUND
Ø3.6 m VEHICLE "ASTRAPE-HV", 23 m · 15 t (MAX GROSS 5 t) · FULL-HULL SLEEVE
Ø3.2 m × 8 m PAYLOAD BAY, CRYO-CLASS FITS WITH MARGIN
PRICE OF WIDE: PLASMA WINDOW POWER SCALES WITH AREA.
Ø4.0 m = 2.8x THE Ø2.4 m BASELINE: CHEAP IN ROCK, DEAR AT MUZZLE.

RING SECTOR ZOOM, ONE OF 24, UNROLLED FLAT

3 PHASE PROPULSION COILS (C = 1.2 m, HOOPS @ 0.6 m)
GAP 200 mm, VACUUM, NO CONTACT
NULL-FLUX LEVITATION LOOPS

VEHICLE REACTION SLEEVE (MAGNET + CONDUCTOR), FULL HULL

PASSIVE PALETTE: NO DRIVE + VEHICLE SETTLES ONTO SLOID BALLS AT BORE BOTTOM
EACH SECTOR: OWN CRYOSTAT BRANCH · OWN QUENCH DUMP · INDIVIDUALLY SWITCHED

STATOR MAGNET MODULE, ONE COIL

VACUUM CASE 316L · NI1 + 80 K SHIELD
NI REBO DOUBLE-PACKAGES @ 20 K
VPI EPOXY · 316L FORMER · SPARK-CLASS BUILD
ONE LOOP · ONE CRYO BRANCH · ONE QUENCH DUMP

FIELD CLASS 5-10 T AT THE GAP, A LAUNCH LSM NEEDS FORCE, NOT FUSION-GRADE FIELD

VACUUM RINGS @ 2.5 m · SEGMENT ELEVATION, 12.5 m UNIT, A SOLID WALL OF MAGNETS EVERY 0.6 m · 20 K + 80 K HEADERS +

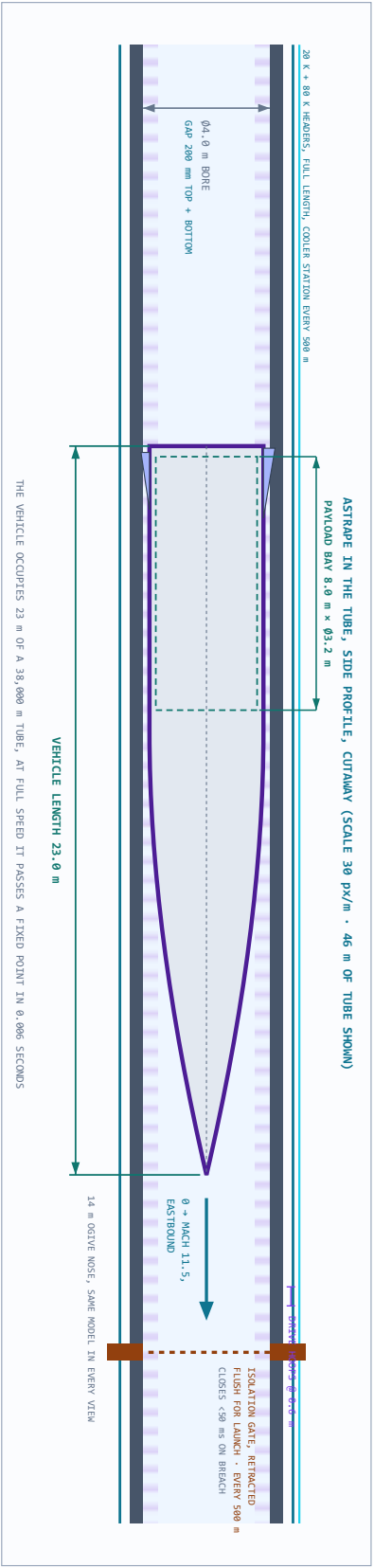
DRIVE HOOPS @ 0.6 m, ~21 PER SEGMENT, ~63,000 ON THE FULL LINE · ALIGNMENT JACKS EVERY ~5 m
CRYO FEED STUB (TOP), COOLER STATION EVERY 500 m · 20 K + 80 K HEADERS RUN THE FULL LENGTH
ISOLATION GATE @ 500 m, BREACH VENTS 500 m, NOT 21.3 km · BELLINGS

PANEL D CONCEPT ONLY, KER-BRO-TUN-001 FOLDED INTO THIS SHEET EVERY DIMENSION SPECULATIVE

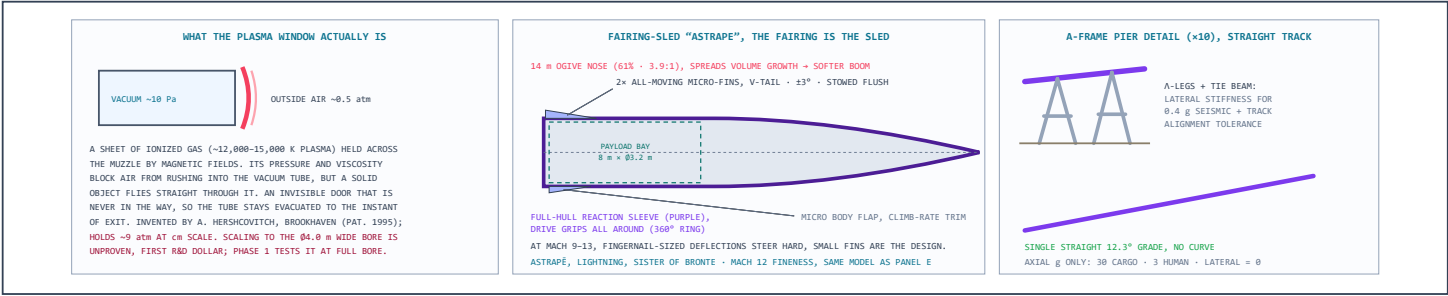
PANEL E, EXTERNAL (BRIDGE) SECTION · ASTRAPE FRONT VIEW · ASTRAPE IN THE TUBE, SIDE PROFILE

THE SAME TUBE, OUTSIDE THE MOUNTAIN, AND THE VEHICLE SHOWN IN IT, FRONT AND SIDE

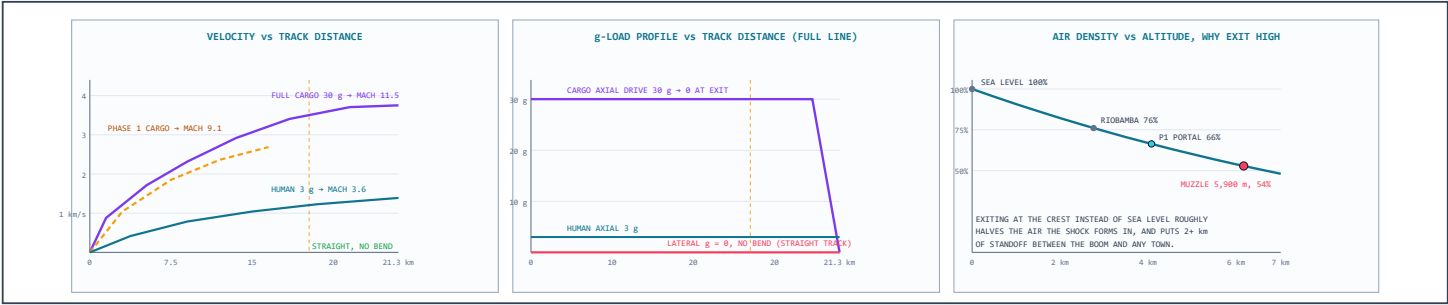
LEFT: VIADUCT ("BRIDGE") SECTION ON A-FRAME PIER · RIGHT: ASTRAPE FRONT VIEW IN THE BORE · BOTTOM: SIDE PROFILE WITH LENGTH DIMENSIONS



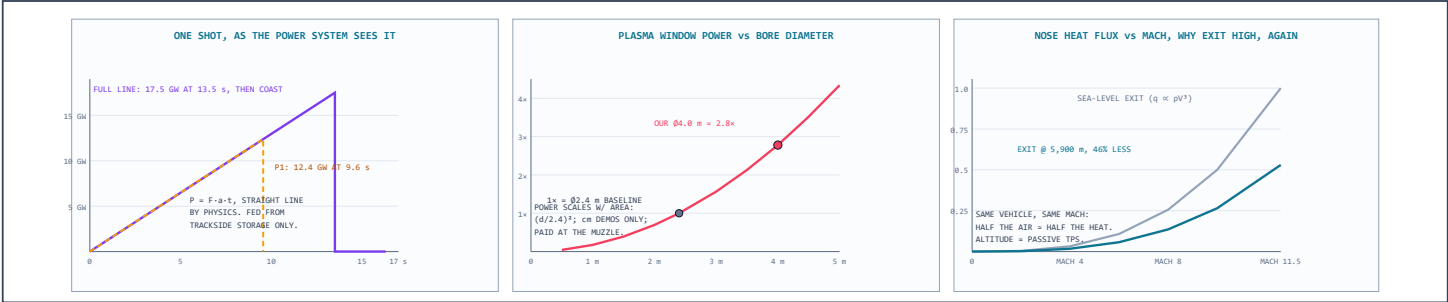
DETAILS, SYSTEMS



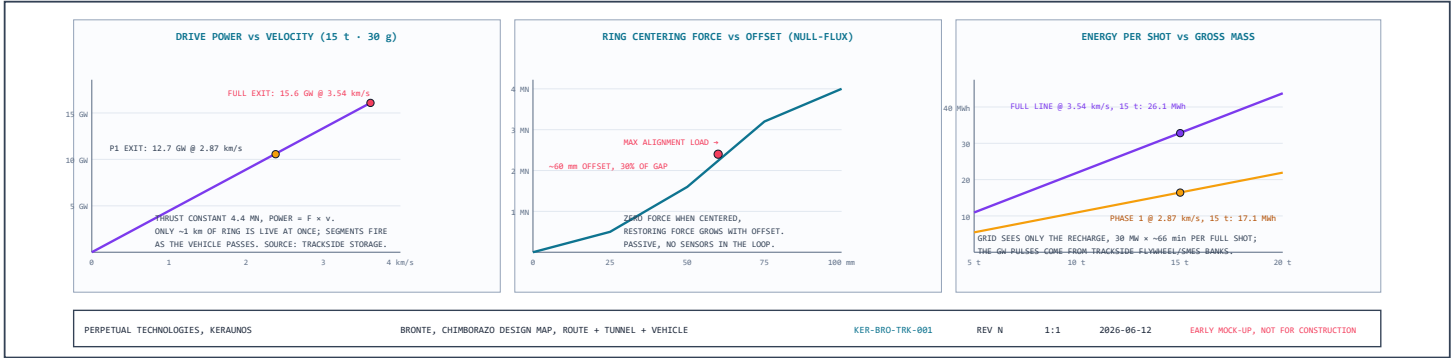
DETAILS, ROUTE ENGINEERING CHARTS



DETAILS, SYSTEMS CHARTS



DETAILS, TUNNEL ENGINEERING CHARTS



REV N, catwalks deleted (the tube is sealed; service is robotic, so nothing for people to stand on), aviation-light label moved off the panel title, and a third charts row added for the support systems. Prior pass: max tonnage set (20 t gross ceiling, 15 t baseline), tube-transit times added for 3/5/30 g, 360° radial truss bracing between vessel and borehole liner, vacuum-jacketed cryo conduit shown for open-air bridge spans, overlaps cleaned, and the ogive softened a touch. Prior note: One vehicle model everywhere now: 23 m total, a 14 m tapering ogive needle nose (3.9:1 nose fineness, stretched specifically to spread the volume growth and soften the sonic boom; it cuts wave drag too) on a 9 m full-diameter cylinder (the math forces the cylinder: an 8 m × Ø3.2 m bay can only pack into the full-diameter aft section, which is exactly 9 m long). Vacuum isolation gates every 500 m sectionalize the tube, retracted flush for launch, closing in under 50 ms on a breach, so a puncture vents 500 m of tube instead of 21.3 km; one is shown retracted in the cutaway. Cryogenics grew to a real system: paired 20 K supply and return headers plus an 80 K shield loop running the full length, cooler stations every 500 m, and invert return lines in the section view. The HVDC bus ducts doubled up and sized up. Earlier revision notes follow. Panel E: the external ("bridge") section showing the identical vessel-and-ring stack riding an A-frame pier in open air with weather cladding instead of rock; ASTRAPE's front view in the bore with the V-tail fins stowed flush (they deploy only after exit, nothing breaks the 200 mm gap envelope in-bore); and the side-profile cutaway placing the full 18 m vehicle inside the tube with explicit dimensions, 18.0 m length, 8.0 m × Ø3.2 m payload bay, Ø4.0 m bore, against the 0.6 m drive-hoop pitch. The payload diameter now carries explicit dimension arrows in every section view. The tunnel section (formerly its own drawing) now lives as Panel D, rebuilt around your call: the drive is a **full 360° stator ring**, 24 sectors per ring, rings every 0.6 m, roughly 37,000 coil hoops on the full line. The line is a single straight 12.3° grade, no curve, so there is no centripetal side-load at all; the ring stays full-circumference for **distributed full-hull thrust** (the pod never builds the axial compression that would buckle a tail-pushed projectile) and precise magnetic centering. The ring levitates, centers, and propels from every direction at once; the vehicle answers with a full-hull reaction sleeve (ASTRAPE's inset updated to match, no belly skid, nothing jettisoned). The bore tightened from Ø5.0 to Ø4.0 m for proper 200 mm magnetic coupling all round, your Ø3.2 m payload is untouched, and the plasma window bill actually improves (2.8× baseline instead of 4.3×). New tunnel charts at the bottom: drive power vs velocity (11–16 GW pulses, only ~1 km of ring live at a time), the null-flux centering curve showing the vehicle riding ~60 mm off-center against alignment loads with no sensors in the loop, and energy-per-shot vs mass with the grid only ever seeing a quiet sub-hour recharge.